```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Npcap Loopback Adapter:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b923:c6f7:7122:8700%6
   Autoconfiguration IPv4 Address. . : 169.254.158.128
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::390c:dc1c:3ca7:94bc%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.10.86
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Windows\system32>_
```

```
C:\Windows\system32>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

C:\Windows\system32>
```

5. netStat : (n
a variety of
active TCP/I
line tool
both incomin
eg: ne

6. nslookup : (n
Perform DN
display DNS
of a pa
for a dom
nslookup
and non

eg:

7. Pathping:
is basicall
Tracert
route to
25 second
the way
of data

5. netstat : (network statistics) netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool of monitoring network connections both incoming and outgoing.

eg: netstat -n.

6. nslookup : (name server lookup is a tool used to Perform DNS lookups in Linux. It is used to display DNS details. Such as the IP address of a particular computer the MX records for a domain of the NS server of a domain nslookup can operate in two modes : interactive and non Interactive

eg: nslookup www.google.com

7. pathping: pathping is unique to windows and is basically a combination of the ping and Tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each other. route along the way, gathering statistics on the route of data

```
C:\Windows\system32>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.

C:\Windows\system32>
```

```
If Dest contains * or ?, it is treated as a shell p
matching destination routes are printed. The '*' matc
and '?' matches any one char. Examples: 157.*.1, 157.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (D
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.
             The route addition failed: The specifie

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*
                                    .... Only prints t
    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.8
              destination^    ^mask          ^gateway

      If IF is not given, it tries to find the bes
      Gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.5

      CHANGE is used to modify gateway and/or met

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32

:\Windows\system32>
```

```
C:\Windows\system32>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  Gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omit
```

Result

The network

successfully

[redacted]

Result

The networking commands were successfully executed and studied